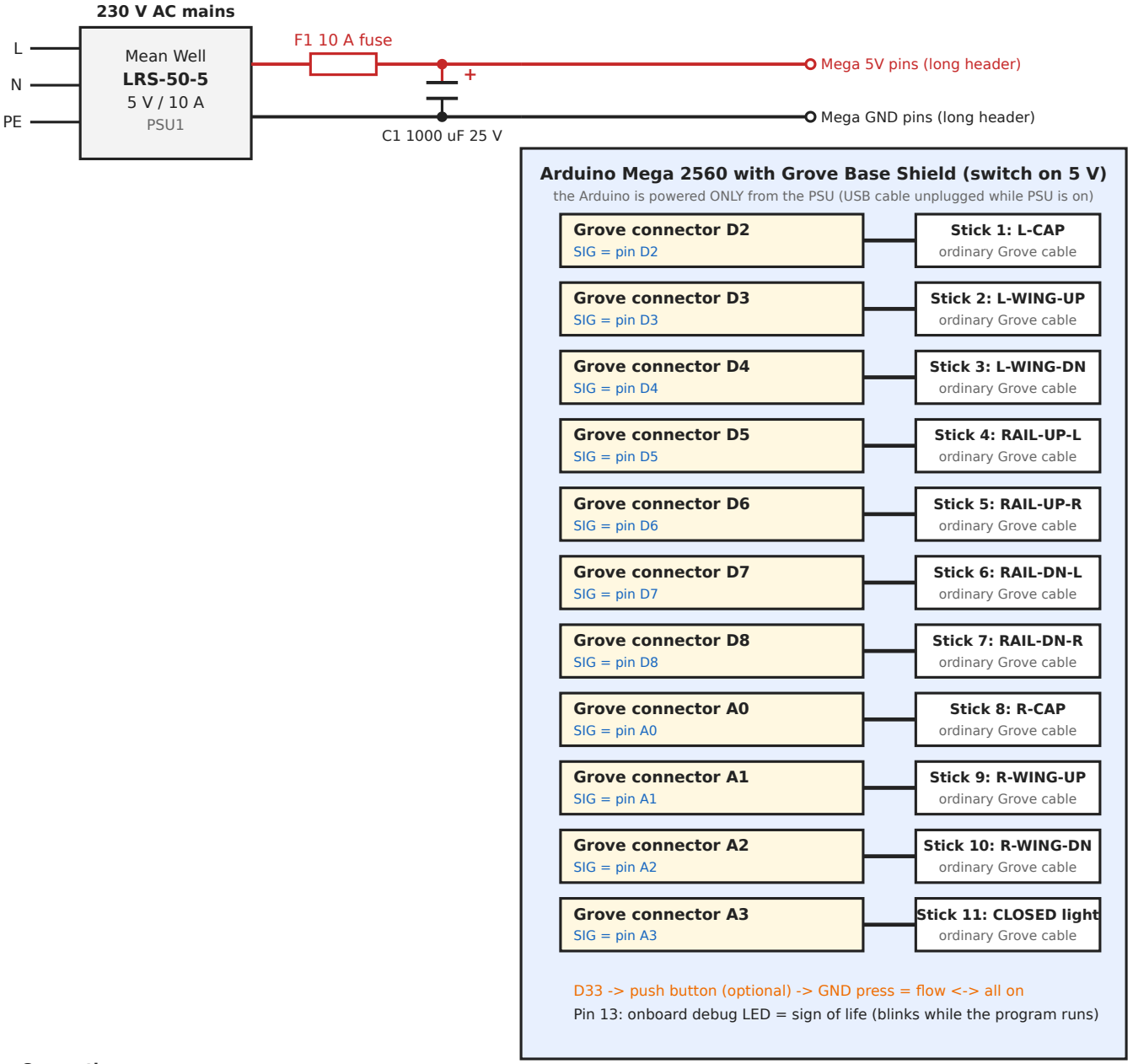
11 sticks - wiring (10 arrow sticks + the CLOSED light stick in the Grove Base Shield)



Connections

1. Each stick is plugged with its own unmodified Grove cable into the n-th connector: stick 1 D2 ... stick 7 D8, stick 8 A0, 9 A1, 10 A2 (the 10 arrow sticks, as in the sign) and stick 11 = the CLOSED light in A3.
2. The shield 3V3/5V switch must be on 5 V. The stick power (red/black wires) goes through the shield and the Grove connectors.
3. PSU V+ -> 10 A fuse F1 -> the Mega 5V pins; PSU V- -> the Mega GND pins (use the 5V and GND pins at the end of the long double header, away from the shield; use both 5V pins and both GND pins, thick wire 1.5 mm2). C1 across the supply wires.
4. Power the Arduino ONLY from the PSU: do not plug the USB cable while the PSU is on (upload the sketch first with USB, unplug, then power).
5. No Serial Monitor output without USB; the pin 13 LED shows that the program runs.
6. Let it run and feel the shield, the connectors, the sticks and the wires now and then; also try the all-on mode (button). If anything gets hot, switch off.
7. The mains side of the supply must be in an enclosure and wired by a person who knows how (nothing here is mains).